[ARCHIVED CATALOGUE]

Financial Engineering (MS)

The objective of this program is the training of graduate students with engineering, applied mathematics or physics backgrounds in the application of mathematical and engineering tools to finance. Financial engineering is a multidisciplinary education program that involves the Viterbi School of Engineering, the USC Marshall School of Business and the USC Dornsife College of Letters, Arts and Sciences (Department of Economics). Financial engineering uses tools from finance and economics, engineering, applied mathematics and statistics to address problems such as derivative securities valuation, strategic planning and dynamic investment strategies, and risk management, which are of interest to investment and commercial banks, trading companies, hedge funds, insurance companies, corporate risk managers and regulatory agencies.

A minimum grade point average of 3.0 must be earned on all course work applied toward the master's degree in financial engineering. Transfer units count as credit (CR) toward the master's degree and are not computed in the grade point average. In addition to the general requirements of the Viterbi School of Engineering, the Master of Science in Financial Engineering is also subject to the following requirements: (1) a total of at least 33 units is required; (2) every plan of study requires prior written approval by the contact faculty of the program; (3) units to be transferred (maximum of four with adviser approval) must have been taken prior to taking classes at USC; interruption of residency is not allowed.

For Admission Requirements, refer to USC Viterbi School of Engineering.

Curriculum

The degree requirements include six required courses and three courses from two lists of electives for a minimum total of 33 units.

Required

- GSBA 548 Corporate Finance Units: 2, 3
  (Note: GSBA 548 for 3 units satisfies this requirement. Do not register for the 2 unit version.)

- ISE 563 Financial Engineering Units: 4
- EE 503 Probability for Electrical and Computer Engineers Units: 4
- EE 512 Stochastic Processes for Financial Engineering Units: 4
- EE 518 Mathematics and Tools for Financial Engineering Units: 4
- EE 590 Directed Research Units: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12
  (1 unit required; maximum 2 units)

Electives (Adviser-Approved)

Finance, Business, Economics Area:

Three courses (9 units) from the following:

- FBE 524 Money and Capital Markets Units: 3
- FBE 529 Financial Analysis and Valuation Units: 3
- FBE 535 Applied Finance in Fixed Income Securities Units: 1.5, 3
- FBE 540 Hedge Funds Units: 3

- FBE 543 Forecasting and Risk Analysis Units: 3
- FBE 551 Quantitative Investing Units: 3
- FBE 554 Trading and Exchanges Units: 3
- FBE 555 Investment Analysis and Portfolio Management Units: 3
- FBE 589 Mortgages and Mortgage-Backed Securities and Markets Units: 3

Optimization, Simulations, Stochastic Systems Area:

One course (4 units) from the following:

- CSCI 567 Machine Learning Units: 4
  or
- DSCI 552 Machine Learning for Data Science Units: 4
  or
- ISE 529 Predictive Analytics Units: 4

- ISE 537 Financial Analytics Units: 4
- ISE 539 Stochastic Elements of Simulation Units: 4
- ISE 530 Optimization Methods for Analytics Units: 4

Units required: 33